

Query Justification

Query 1: Average Grade per Subject

Business Goal: Curricular Assessment & Teacher Support.

- Logic: The school administration needs to evaluate the difficulty level of various subjects and the effectiveness of the current curriculum. By identifying subjects with significantly lower averages, the school can allocate additional resources or training to those departments.
- How it works: We calculate the average grade for all students enrolled in a specific subject. This query also includes a count of students per subject to ensure the average is statistically significant (e.g., an average of 90 in a class of 1 student is less informative than in a class of 30).

Query 2: Daily Attendance Summary

Business Goal: Operational Logistics & Student Safety.

- Logic: For daily operations, the school must know the real-time status of the student body. High "Late" counts might indicate transportation issues, while high "Absent" counts could signal a health outbreak or safety concern.
- How it works: This query provides a summary of the current day's attendance records. It groups the student population into three categories: 'Present', 'Absent', and 'Late', providing a quick quantitative dashboard for the morning assembly or administrative check.

Query 3: Class Performance and Teacher Report

Business Goal: Cross-Departmental Visibility.

- Logic: In a large school, it is difficult for administrators to keep track of which teachers are responsible for which students across different floors and subjects. This query creates a unified view that connects physical locations (Classrooms) with academic leaders (Teachers).
- How it works: This is a multi-table join that links students to their physical classroom floor and their subject-specific teachers. It provides a "master list" that helps staff

identify who to contact regarding a student's performance in a specific area of the building.

Query 4: Parent-Teacher Communication Bridge

Business Goal: Stakeholder Engagement & Relationship Management.

- Logic: Effective communication is the backbone of a school. When a student struggles or excels, the school needs an immediate way to pull the contact details for both the legal guardians (Parents) and the academic staff (Teachers) assigned to that specific student.
- How it works: We link the parent-student relationship table with the teacher-classroom assignment table. This isolates a specific student and returns a direct mapping of their parents' usernames alongside their teachers' names, facilitating rapid communication for parent-teacher conferences.